Western Governor's University

D209 Data Mining

Task 2: Random Forest Regression

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Part I: Research Question

A. Describe the purpose of this data mining report by doing the following:

1. Propose **one** question relevant to a real-world organizational situation that you will answer using **one** of the following prediction methods:

   Is it possible to predict the number of initial days spent in the hospital based on known variables in the medical dataset using the random forest method?

2. Define **one** goal of the data analysis. Ensure that your goal is reasonable within the scope of the scenario and is represented in the available data.

   One goal of this data analysis is to build a random forest model capable of providing meaningful predictions of how many days patients will need to stay in the hospital to allow for the hospital to prepare accordingly.

Part II: Method Justification

B. Explain the reasons for your chosen prediction method from part A1 by doing the following:

1. Explain how the prediction method you chose analyzes the selected data set. Include expected outcomes.

   The random forest model uses an average of multiple decision trees, providing a statistically robust form of prediction, hopefully ensuring high accuracy. This combination of multiple decision trees for an output is called ensemble method. Specifically, the type of ensemble method used for random forest regression is called bagging. Bagging works by randomly sampling the dataset to create multiple prediction models. These individual models are called bootstrap samples and use row sampling to attain the data to build the model. This is done using a process called bootstrapping, meaning rows are sampled with replacement. Now the model is trained, creating parallel decision trees which will run simultaneously and then aggregate their outputs into a random forest prediction. (Sruthi)

2. Summarize **one** assumption of the chosen prediction method.

   Random forest models assume that the randomly sampled data is accurately representative of the full dataset. (Sruthi)

3. List the packages or libraries you have chosen for Python or R and justify how *each* item on the list supports the analysis.

Pandas – This allows transformation of the dataset from a csv into a dataframe that can be manipulated and analyzed.

Numpy – Numpy allows for manipulation on data series, which will be necessary to calculate model metrics.

sklearn.model_selection import train_test_split - This package is necessary in order to create and analyze a testing data set.

from sklearn.metrics import mean_squared_error as MSE – This will allow for simple calculations for MSE on the model.

from sklearn.ensemble import RandomForestRegressor – This contains the RFR model used for this analysis.

from sklearn.metrics import r2_score as r2 – This will allow for R squared calculations on the model.

from sklearn.model_selection import GridSearchCV - This will allow for fitting and scoring of the RFR model onto the dataset.

Seaborn – This is used to graph data distribution for outlier analysis.

from sklearn import tree – Used to visualize and customize decision tree.

Matplotlib – Used to implement a heatmap for multicollinearity assessment.

Part III: Data Preparation

C. Perform data preparation for the chosen data set by doing the following:

1. Describe **one** data preprocessing goal relevant to the prediction method from part A1.

The medical dataset will need to be converted to dummies and then one hot encoded before processing the model.

2. Identify the initial data set variables that you will use to perform the analysis for the prediction question from part A1 and group *each* variable as numeric or categorical.

Numeric: Income, TotalCharge, Age, VitD_levels, Initial days
Categorical: Initial_admin, ReAdmis, Complication_risk, services, stroke, asthma, diabetes, overweight, highblood,

3. Explain the steps used to prepare the data for the analysis. Identify the code segment for *each* step.

Lines 13-14: First, the data is evaluated for duplicate entries using .duplicated, none are found.
Lines 16-36: Then, all categories in the dataset are evaluated to assess unique values, no issues are discovered.
Lines 38-39: A .dropna function is run over the entire dataframe to ensure any invalid values will not hold up calculations.
Lines 41-87: Outliers are evaluated for numeric columns and any outlying values are imputed with the column median.
Lines 92-96: After selecting variables for analysis, the variables are transformed into dummies, and one hot encoded.

Lines 98-99: Then any binary dummy columns will have the second option removed from the dataset, as one dummy column contains all necessary information for binary variables.

Lines 101-104: After the data has been otherwise cleaned, a heatmap is created using the remaining dummy variables to assess multicollinearity. There are no variables with a strong enough correlation to suggest they should be removed.

(Beheshti) (GeeksforGeeks)

4. Provide a copy of the cleaned data set.

See file attached: “D209Proj2CleanedDataSet.csv”

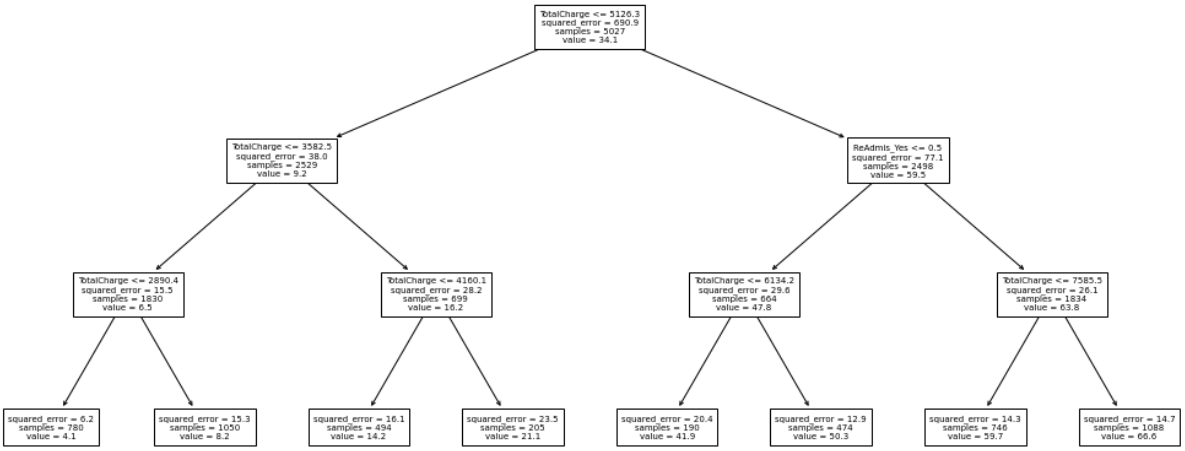
Part IV: Analysis

D. Perform the data analysis and report on the results by doing the following:

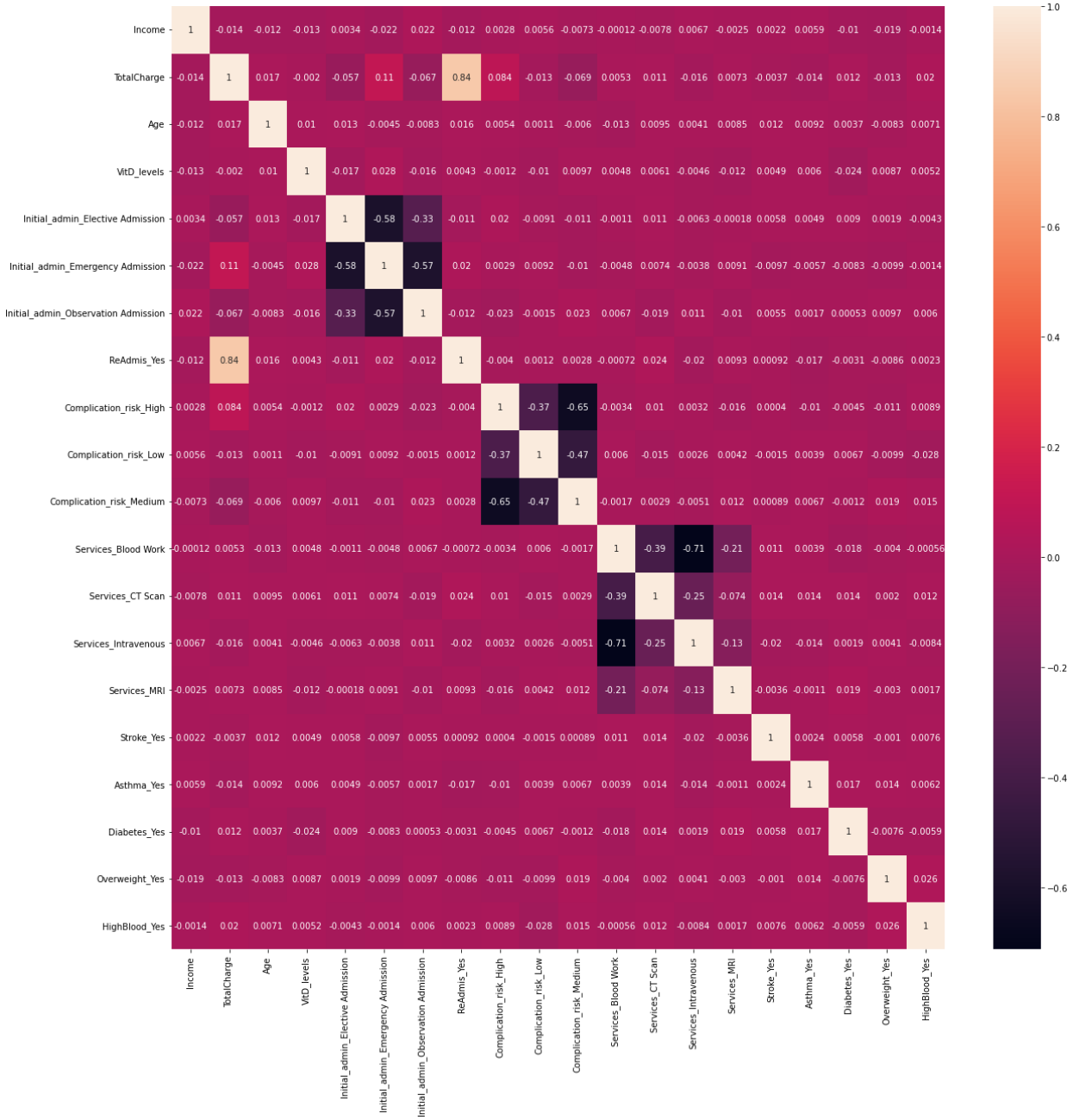
1. Split the data into training and test data sets and provide the file(s).
   See files attached: “X_train.csv”, “X_test.csv”, “y_train.csv”, “y_test.csv”

2. Describe the analysis technique you used to appropriately analyze the data. Include screenshots of the intermediate calculations you performed.

   Random forest regression analyzes the data set by taking multiple subsets of the data and making a prediction using each subset. Then, all subset predictions are averaged to create a final prediction the model will provide. (GeeksforGeeks) (Beheshti)



(Sklearn.tree.plot_tree)


Income
TotalCharge
Age
VitD_levels
Initial_admin_Elective Admission
Initial_admin_Emergency Admission
Initial_admin_Observation Admission
ReAdmis_Yes
Complication_risk_High
Complication_risk_Low
Complication_risk_Medium
Services_Blood Work
Services_CT Scan
Services_Intravenous
Services_MRI
Stroke_Yes
Asthma_Yes
Diabetes_Yes
Overweight_Yes
HighBlood_Yes
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6

(Waskom)

```
Out[19]:
GridSearchCV(cv=3,
             estimator=RandomForestRegressor(max_depth=3, random_state=15),
             param_grid={'max_depth': [8, None], 'max_features': [2, 3, 4],
                         'n_estimators': [10, 50, 100]},
             verbose=3)

In [20]:
    ...:
    ...: y_pred = cv.predict(X_test)

In [21]:
    ...:
    ...: print("MSE: ", MSE(y_test,y_pred))
    ...: print("Root MSE: ", MSE(y_test,y_pred)**1/2)
    ...: print("R Square: ", r2(y_test,y_pred))
    ...:
    ...: #for potential model tuning
    ...: print("Best params suggestion: ", cv.best_params_)
    ...:
    ...: print("cv.best score: ",cv.best_score_)
MSE:  5.979904143861663
Root MSE:  2.9899520719308317
R Square:  0.9912199667854843
Best params suggestion:  {'max_depth': None, 'max_features': 4, 'n_estimators': 100}
cv.best score:  0.9895403240882853
```

(SciKit Learn) (Shah)

3. Provide the code used to perform the prediction analysis from part D2.

See code attached: "Brandon Velez D209 Task 2.py"

Part V: Data Summary and Implications

E. Summarize your data analysis by doing the following:

1. Explain the accuracy(metrics) and the mean squared error (MSE) of your prediction model.

Mean squared error represents the averaged squared discrepancy between the predicted y values from the model and the actual y values from the dataset. Therefore, the lower the MSE, the stronger the model fits. The random forest model having a MSE of 5.97 initial days and therefore a Root MSE of 2.98 days shows that the model is, on average, outputting predicted y values that are 2.98 days away from the actual y value during testing. (Behesti)

This model scores an R square of 0.991 shows that 99.1% of the change in the y value, initial days, can be accounted for and predicted using the model features. This is a very strong fit and indicates the model is able to make accurate predictions within 1% of the actual y value. (Viswa)

2. Discuss the results and implications of your prediction analysis.

The MSE and R square of the random forest model are both very strong and indicate that the model is able to provide statistically relevant predictions. This model is accurate enough to be implemented for use in the hospital. Using the .best_params_

function, feature selection could be implemented on this model, further boosting its real world relevance as it could run off less data and less computing power than in this project.

3. Discuss **one** limitation of your data analysis.

One limitation of this data analysis is that the random forest model is built using past data. The collected data is only representative of one hospital's experience over one time period, so this model may have no relevance outside of the range of data it has been trained on. There could be many unknown factors affecting how the data was distributed over the initial dataset, and those factors could easily change on the next dataset analyzed. (GeeksforGeeks)

4. Recommend a course of action for the real-world organizational situation from part A1 based on your results and implications discussed in part E2.

Based on the strong root MSE and R-squared scores, this random forest model should be implemented to give an initial days estimate on patient's stay in real time. The hospital now has access to a relevant prediction for how long patients will need hospital treatment, and should use that data to staff and supply the hospital accordingly. This will be most useful when the hospital experiences uncommonly high patient traffic, and can alleviate a great deal of uncertainty as to whether more staff and supplies will be needed in time to prepare accordingly.

Part VI: Demonstration

F. Panopto Link: https://wgu.hosted.panopto.com/Panopto/Pages/Viewer.aspx?id=45ce9d96-6f46-471c-b8ee-b2cf00ff68b5

G&H:

Third Party Code Sources:

Beheshti, Nima. "Random Forest Regression | towards Data Science." *Towards Data Science*, 2 Mar. 2022, towardsdatascience.com/random-forest-regression-5f605132d19d/.

GeeksforGeeks. "Random Forest Regression in Python." *GeeksforGeeks*, 14 June 2019, www.geeksforgeeks.org/random-forest-regression-in-python/#. Accessed 23 Apr. 2025.

SciKit Learn. "Sklearn.model_selection.GridSearchCV — Scikit-Learn 0.22 Documentation." *Scikit-Learn.org*, 2019, scikit-learn.org/stable/modules/generated/sklearn.model_selection.GridSearchCV.html.

Shah, Rahul. “Tune Hyperparameters with GridSearchCV.” *Analytics Vidhya*, 23 June 2021, www.analyticsvidhya.com/blog/2021/06/tune-hyperparameters-with-gridsearchcv/#h-how-to-apply-gridsearchcv. Accessed 22 Apr. 2025.

“Sklearn.tree.plot_tree.” *Scikit-Learn*, scikit-learn.org/stable/modules/generated/sklearn.tree.plot_tree.html.

Waskom, Michael. “Seaborn.heatmap — Seaborn 0.10.1 Documentation.” *Seaborn.pydata.org*, 2024, seaborn.pydata.org/generated/seaborn.heatmap.html.

Works Cited

Sruthi. “Understanding Random Forest Algorithm with Examples.” *Analytics Vidhya*, 17 June 2021, www.analyticsvidhya.com/blog/2021/06/understanding-random-forest/#h-what-is-random-forest.

Viswa. “Surface Level Understanding of Random Forest Regression.” *Medium*, 15 Aug. 2023, medium.com/@vk.viswa/surface-level-understanding-of-random-forest-regression-3bac37e06f3e.